

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection without human labeling.

Source: arxiv.org/abs/2504.02882

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must control conversation flow: ask follow-up questions, make tool calls, or reject unavailable tools. SFT with expert trajectories alone cannot teach these skills.

Failure Mode 1

Missing Follow-Up Questions

Model skips slot-filling when user query lacks required arguments, leading directly to incomplete tool calls.

Failure Mode 2

Hallucinated Tool Invocations

Model fabricates tool names or argument values that don't exist, producing incorrect results for the user.

Failure Mode 3

Failed Tool Rejection

Model attempts tool calls for out-of-scope requests instead of politely declining when no suitable tool exists.

→ Motivation: Use Direct Preference Optimization (DPO) to learn from paired correct/incorrect dialogue trajectories, teaching the model when to ask, call, or reject — automatically, without human annotation.

Modeling TA-LLM Dialogue as a 5-State MDP



Key Insight: The MDP formulation with 5 internal (unobserved) states enables systematic decomposition of dialogue flow into distinct state trajectories — the foundation for automatic preference pair construction.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	All required arguments present. No slot-filling needed.	"Book a flight from NYC to LAX on March 15 for 2 passengers"
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Missing arguments. Slot-filling QA repeats N times.	"Book a flight" → model asks departure, destination, date...
Type 3	$1 \rightarrow 1$	Out-of-scope request. Tool call must be rejected.	"Write me a poem about cats" → no matching tool available

Type 1: Complete Query

$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Direct tool execution

Type 2: Incomplete Query

$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$

Slot-filling loop required

Type 3: Out-of-Scope

$1 \rightarrow 1$

Rejection, stays in S1

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Prevent redundant slot-filling	2,089	Easy
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Prevent redundant slot-filling	3,092	Hard
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	$1 \rightarrow 1$	Tool call accept	2,652	Mixed
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Prevent slot hallucination	2,651	Mixed
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	$1 \rightarrow (2 \times M) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Prevent slot hallucination	2,530	Hard
Type 3	$1 \rightarrow 1$	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4$	Tool call reject	1,129	Hard

Each pair teaches a specific dialogue control lesson. Chosen trajectories follow correct state transitions; rejected trajectories exhibit common SFT failure modes. No human annotation required.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

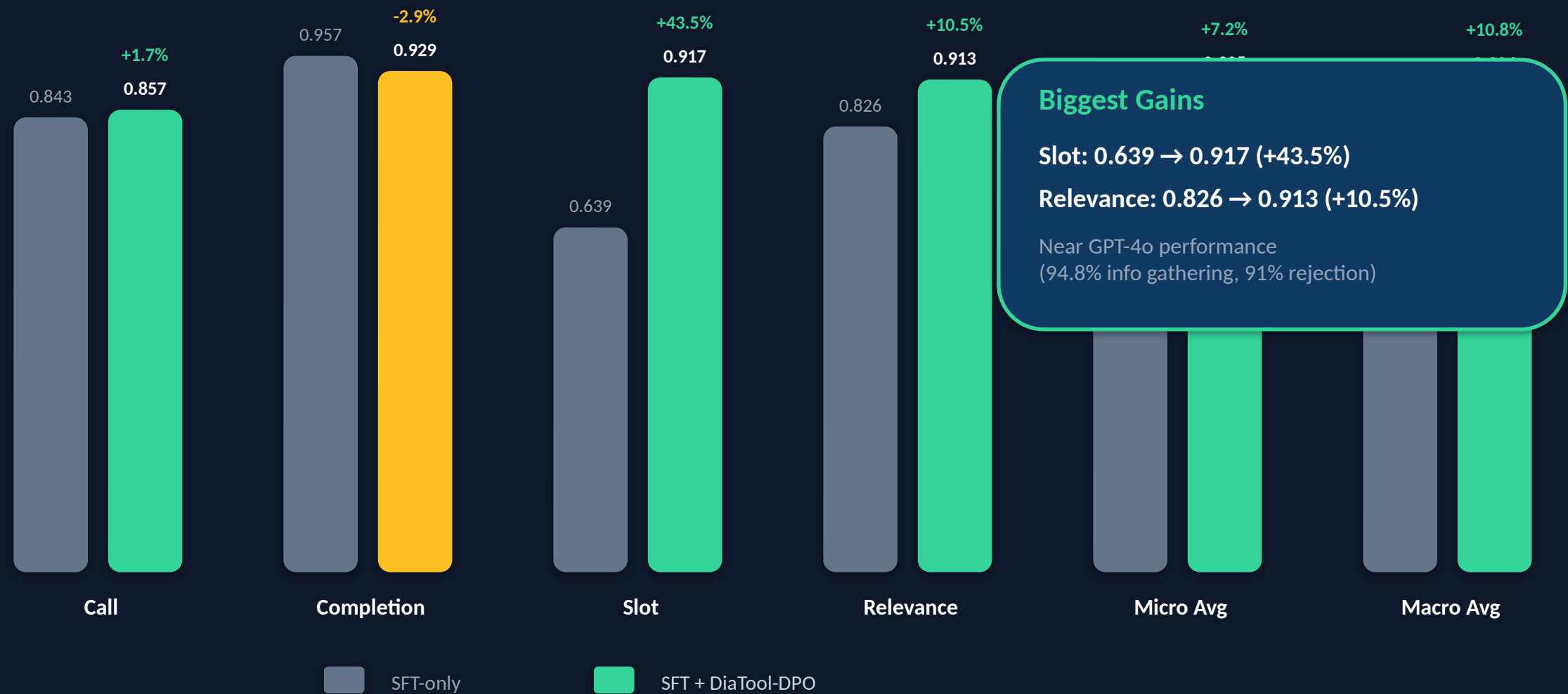
FunctionChat-Bench Evaluation Results

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

Bold green = best in category | * denotes column-best within model group

LLaMA-3-8B: +43.5% Slot Accuracy, +10.5% Relevance

Impact of adding DiaTool-DPO to SFT training



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

MDP Formulation Enables Systematic Pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

DPO Complements SFT — Not a Replacement

DiaTool-DPO alone performs poorly. Combined with SFT, it consistently achieves the best balanced Macro Average across all model sizes.

Largest Gains in Slot-Filling and Tool Rejection

LLaMA-3-8B: Slot accuracy +43.5% (0.639→0.917), Relevance +10.5% (0.826→0.913) — the two skills SFT struggles with most.

Language-Agnostic Design

Primary experiments in Korean, but the method generalizes to English — broad applicability for production tool-augmented LLM systems.

Path to GPT-4o-Level Tool Use on Smaller Models

SFT + DiaTool-DPO achieves near GPT-4o performance (94.8% information gathering, 91% tool call rejection) on open-source 8B models.